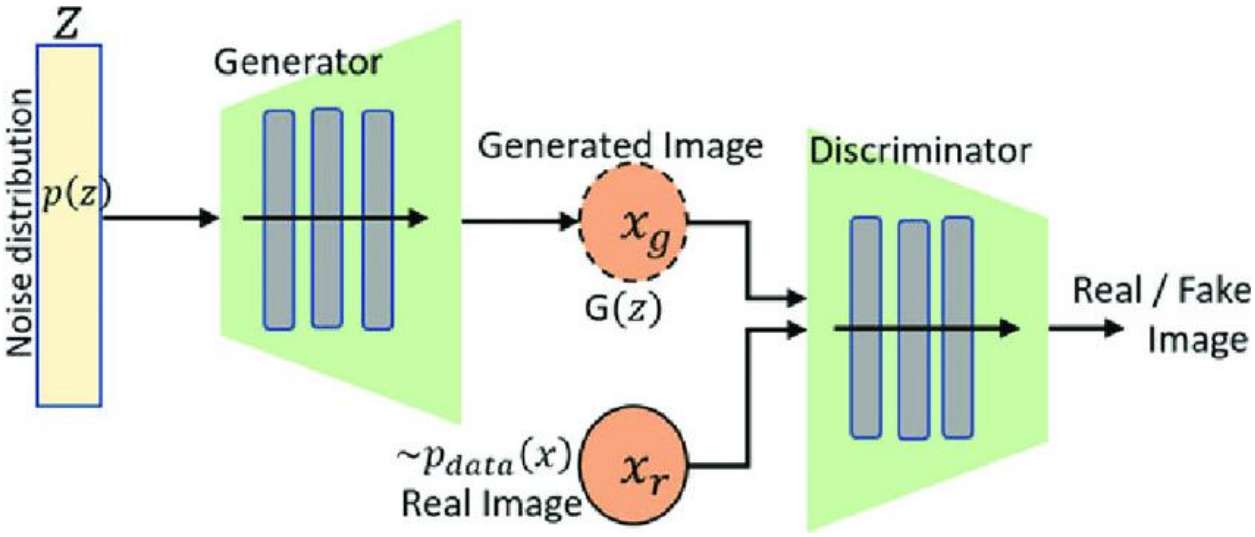
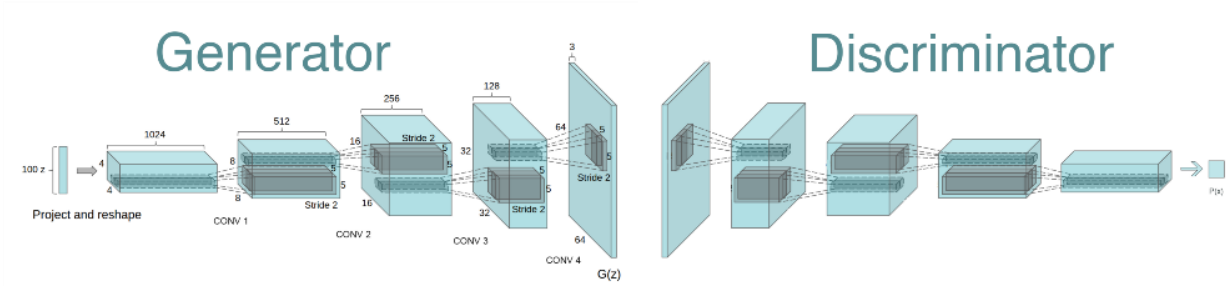
Standard GAN (Vanilla GAN)



GAN(Multi layer perceptron)

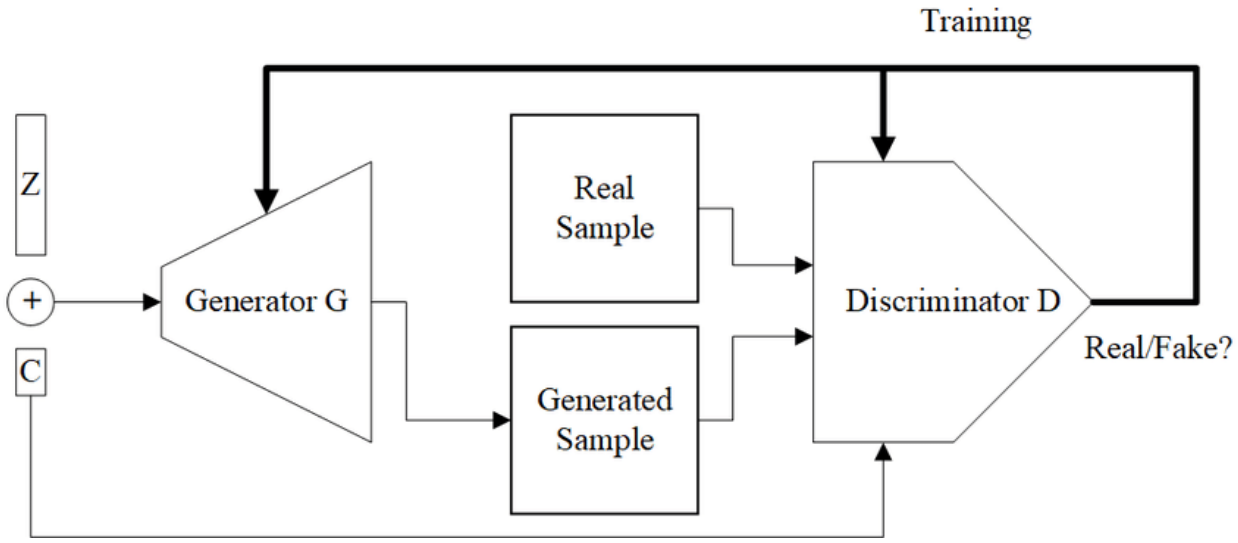
DCGAN

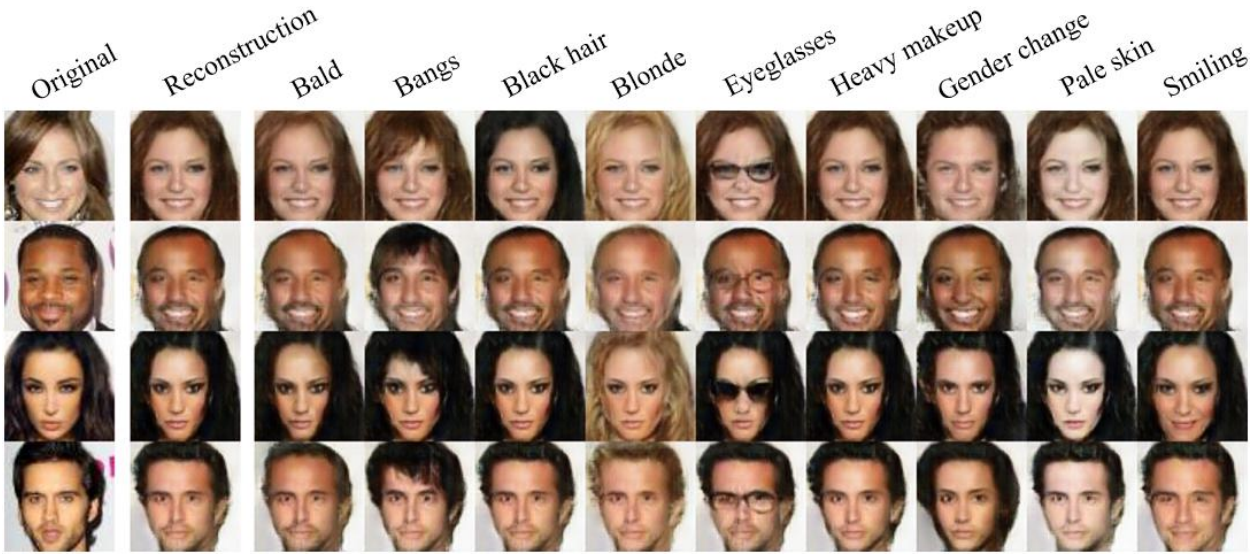
Deep Convolutional GAN (DCGAN)



Conditional GAN (cGAN)

The input noise z and the output of D are conditioned on some extra information, c (e.g., class labels).





Wasserstein GAN (WGAN)

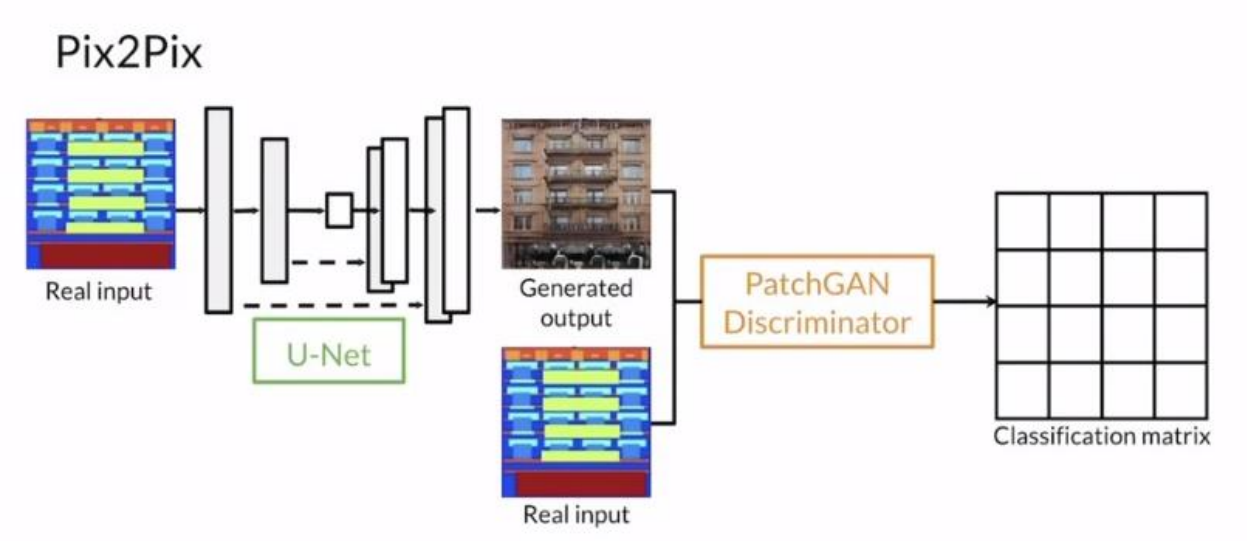
Critic (D) outputs real numbers, not probabilities. Replaced the JS Divergence loss with the **Earth Mover's Distance** (Wasserstein distance)

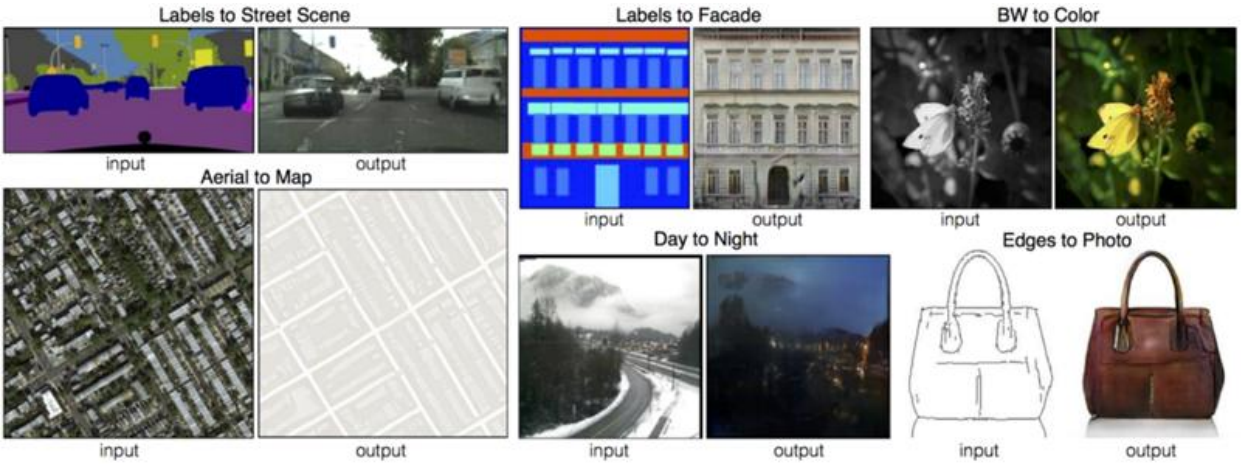
$$\text{GAN: } \nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^{m} \left[log D(\mathbf{x}^{(i)}) + log(1 - D(G(\mathbf{z}^{(i)}))) \right]$$

$$\text{WGAN: } \nabla_{w} \frac{1}{m} \sum_{i=1}^{m} \left[f(\mathbf{x}^{(i)}) - f(G(\mathbf{z}^{(i)})) \right]$$

Pix2Pix (Image-to-Image)

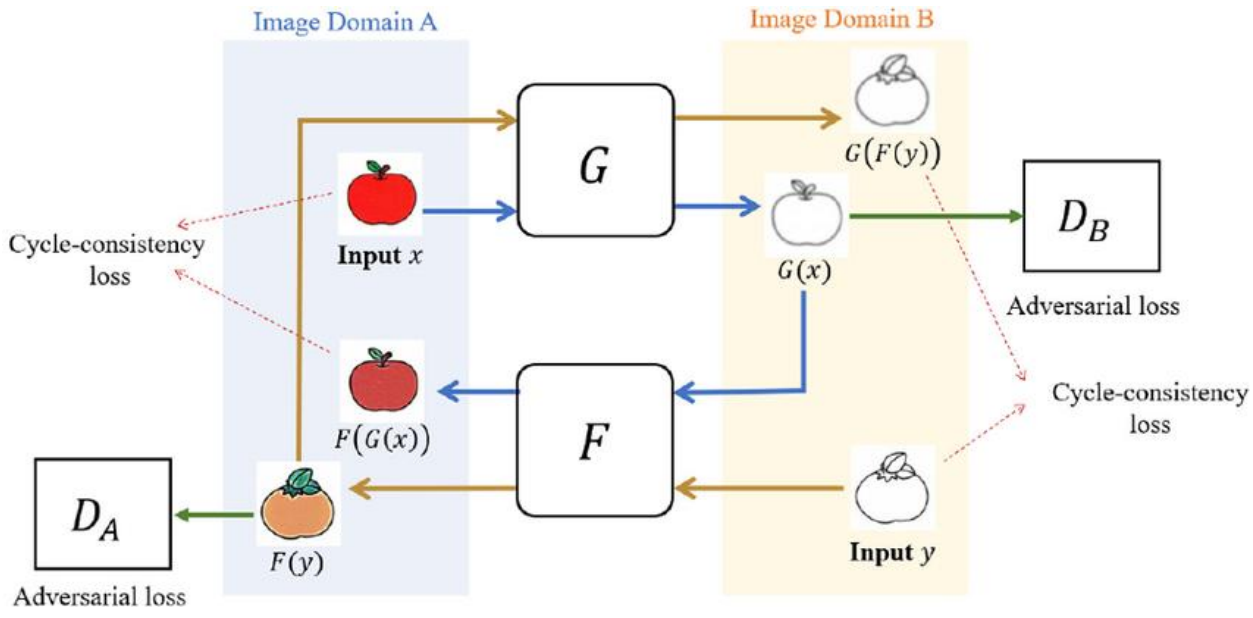
A cGAN that conditions the generation on a full input image, using a U-Net architecture for the Generator

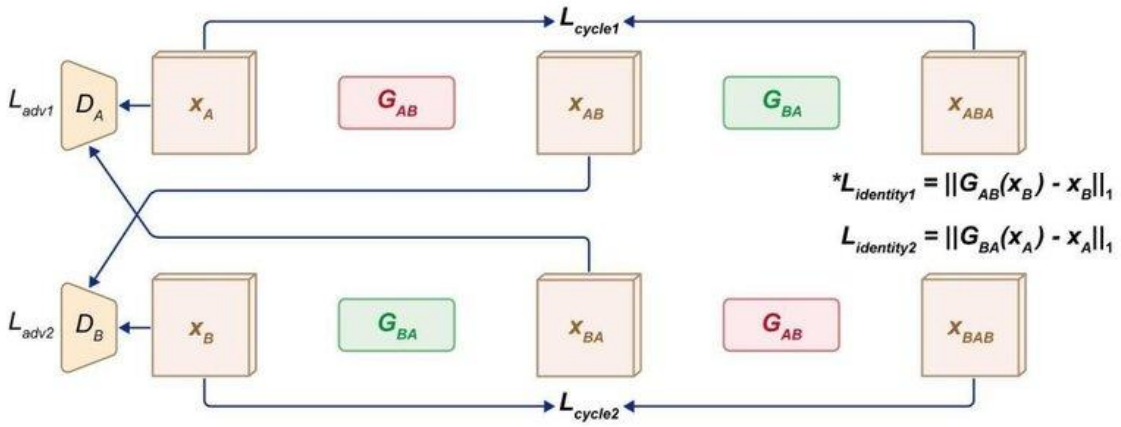


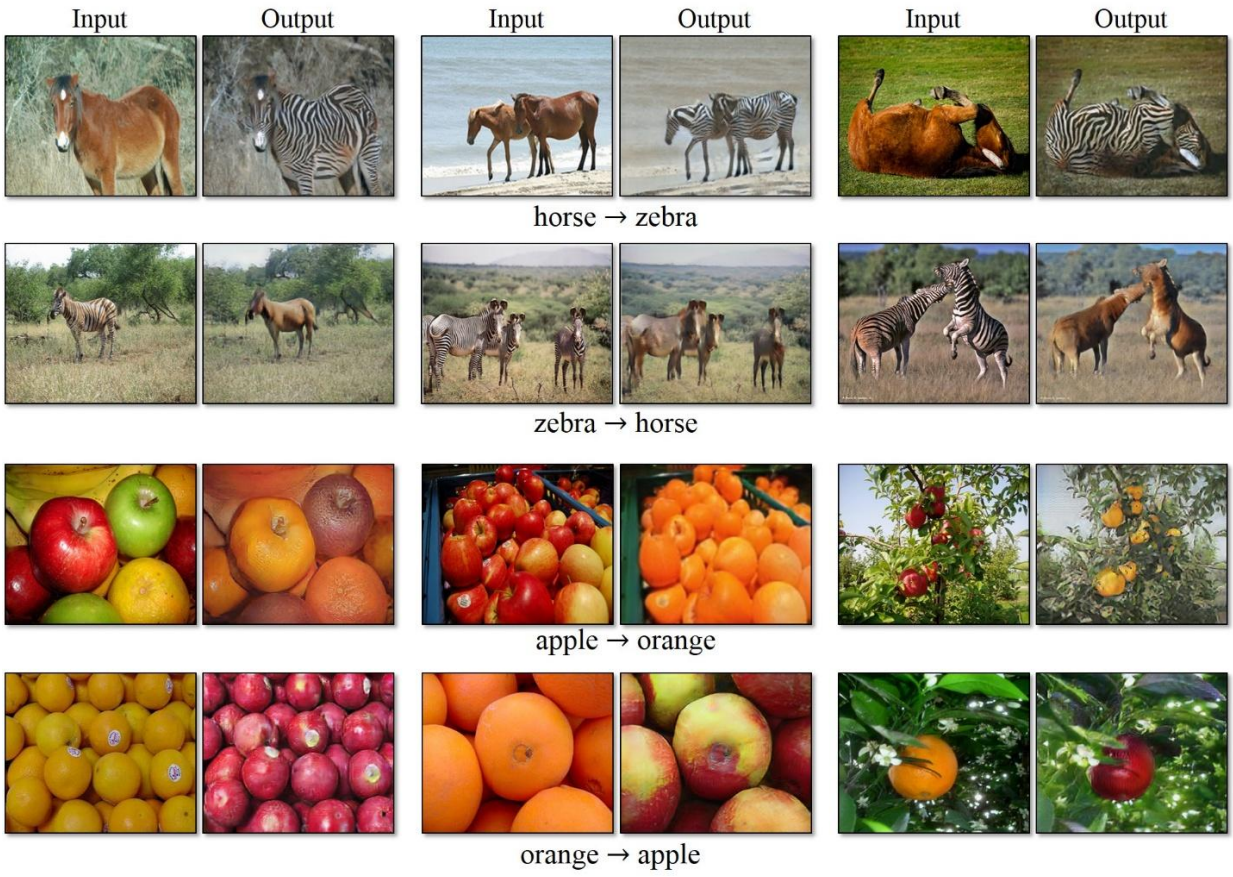


CycleGAN

Uses a cycle-consistency loss to learn the mapping between two domains (A -> B and B -> A) *without* requiring paired training data.

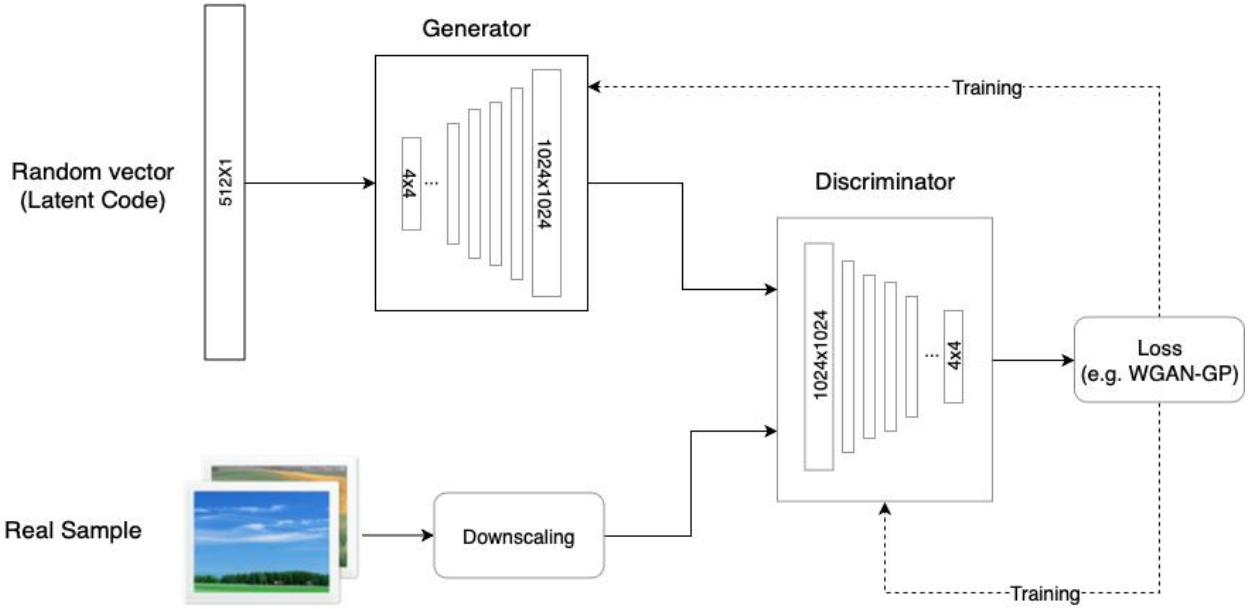


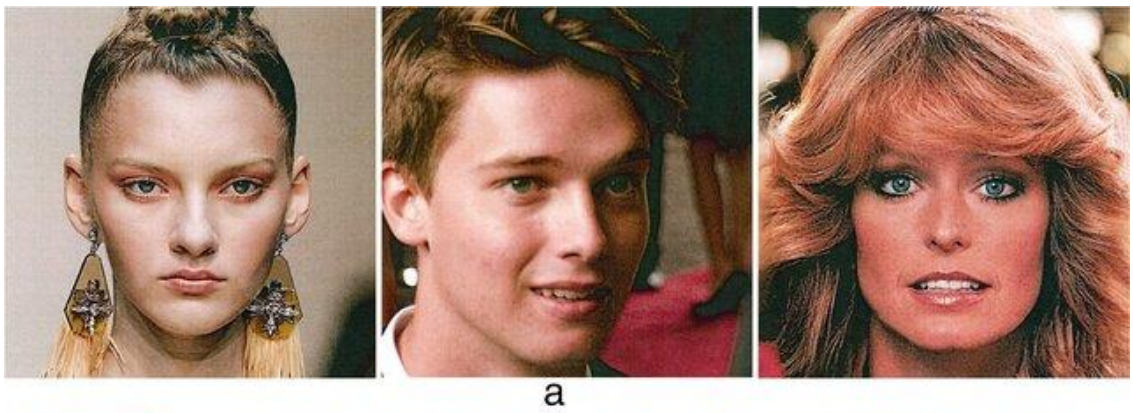



Input
Output
Input
Output
Input
Output
horse → zebra
zebra → horse
apple → orange
orange → apple

ProGAN (Progressive Growing GAN)

Starts training with very low-resolution images and progressively adds new layers to both the G and D as training advances. Uses Bi-Linear Sampling.





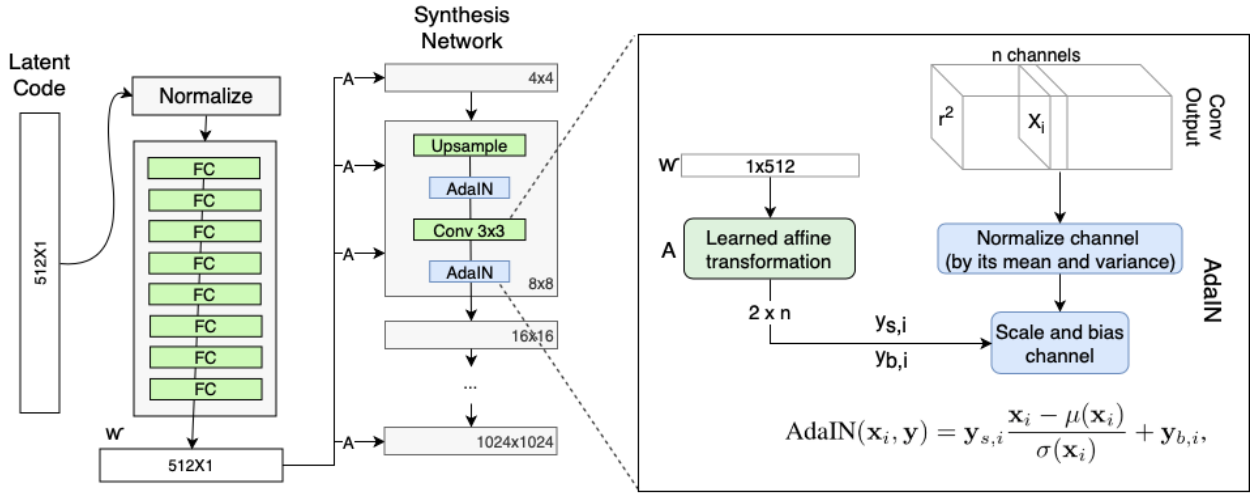
a

b

StyleGAN

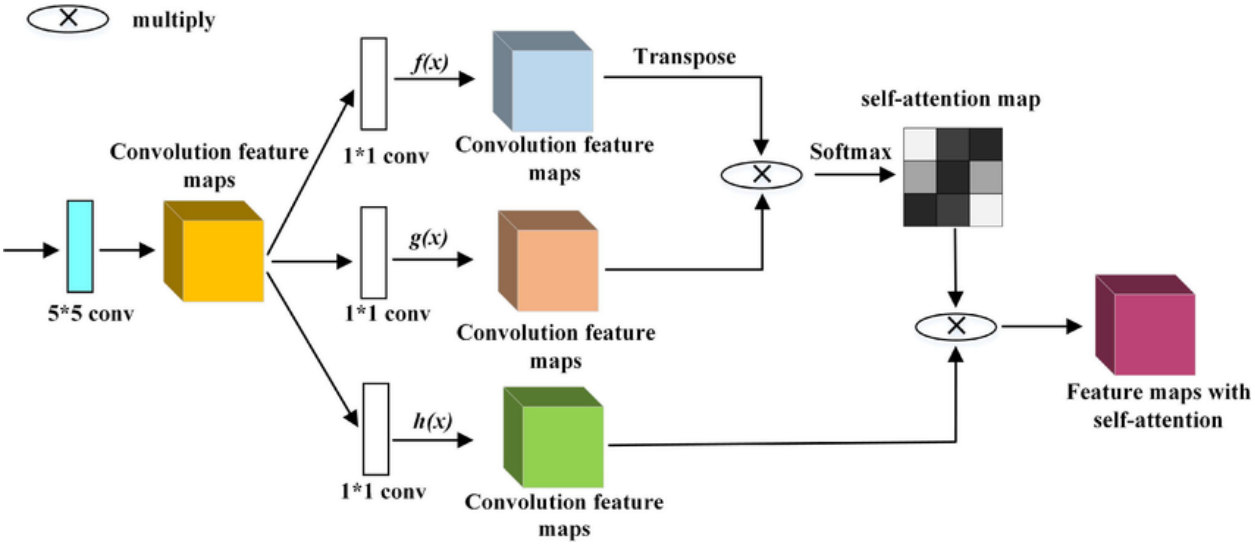
Introduced style-based architecture where latent code is processed by a mapping network and fed into the Generator at multiple scales via adaptive instance normalization (AdaIN). Progressive growing+ Bilinear Sampling+Mapping Network+AdaIN.

Adaptive Instance Normalization aligns the statistics of one feature map to another. StyleGAN uses it to inject style information into each convolution block.



Self-Attention GAN (SAGAN)

Incorporated attention mechanisms into G and D, allowing them to focus on globally consistent details regardless of distance in the image.



StackGAN

Generates high-resolution images from text descriptions by using a multi-stage process: first generating a low-res image, then refining it in subsequent stages.